

Seeing Through Deepfakes: A Human-Inspired Framework for Multi-Face Detection

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Abstract

Multi-face deepfake videos are becoming increasingly prevalent, often appearing in natural social settings that challenge existing detection methods. Most current approaches excel at single-face detection but struggle in multi-face scenarios, due to a lack of awareness of crucial contextual cues. In this work, we develop a novel approach that leverages human cognition to analyze and defend against multi-face deepfake videos. Through a series of human studies, we systematically examine how people detect deepfake faces in social settings. Our quantitative analysis reveals four key cues humans rely on: scene-motion coherence, inter-face appearance compatibility, interpersonal gaze alignment, and face-body consistency. Guided by these insights, we introduce *HICOM*, a novel framework designed to detect every fake face in multi-face scenarios. Extensive experiments on benchmark datasets show that *HICOM* improves average accuracy by 3.3% in in-dataset detection and 2.8% under real-world perturbations. Moreover, it outperforms existing methods by 5.8% on unseen datasets, demonstrating the generalization of human-inspired cues. *HICOM* further enhances interpretability by incorporating an LLM to provide human-readable explanations, making detection results more transparent and convincing. Our work sheds light on involving human factors to enhance defense against deepfakes.

1. Introduction

The rapid rise of AI-generated content has made it easier to create and spread fake videos featuring multiple altered faces, increasing the risk of public manipulation and harm [77]. Since humans naturally interact in groups, detecting fake faces in multi-face social settings is especially critical. For instance, a recent news report [56] revealed how fraudsters used fake faces of a CFO and employees in a video group meeting to deceive and defraud HK\$25 million. This case underscores the urgent need for detection methods that

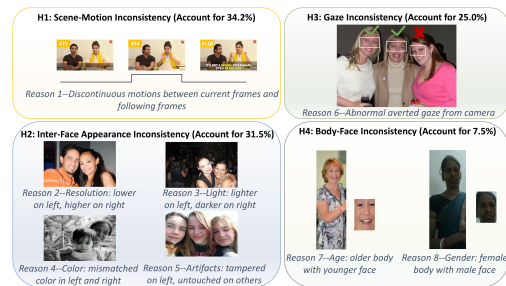


Figure 1. This work takes a novel human-centric approach to multi-face deepfake detection. We conducted a series of human studies to test four research hypotheses (H1-H4), identifying eight cues that humans rely on to detect deepfake faces. These cues later informed the design of our computational model, *HICOM*, which detects every deepfake face in multi-face scenarios.

account for group dynamics and contextual cues to prevent such deceptive practices.

Accurately detecting every fake face within a video frame is crucial, as understanding a scene depends on the interplay among all faces present. For example, a deepfake video circulating on social media falsely depicts U.S. President Donald Trump, Vice President J.D. Vance, and Ukrainian President Volodymyr Zelenskyy engaging in a physical altercation inside the White House following their February 2025 dispute [6]. The video, which gained traction on TikTok [72], misrepresents the nature of the confrontation. If only Trump’s manipulated face is flagged as fake while Zelenskyy’s and Vance’s faces are mistakenly classified as genuine, viewers may still believe the fight occurred, albeit with Zelenskyy and Vance only. This highlights the need for comprehensive, frame-level complete multi-face detection¹ in social settings.

However, *multi-face deepfake detection is an emerging scenario that is inherently more complex than traditional single-face detection, introducing new challenges*. Unlike single-face scenarios—where methods focus solely on in-

¹Frame-level complete multi-face detection considers a frame accurate only if every single face within the frame is correctly classified.

dividual facial characteristics—multi-face scenarios require understanding how multiple faces interact within the same scene. Simply applying single-face detection methods independently to each face in a multi-face setting is insufficient because it ignores critical contextual information about the scene-motion relationships and interactions among faces [43, 81]. In multi-face settings, faces often engage in visual interactions that introduce perceptual cues—such as temporal correlation, coherence in visual attributes, and scene consistency—that are essential for accurate interpretation.

Challenges. *The deepfake detection community currently lacks a well-established benchmark for frame-level complete multi-face detection.* Existing methods focus mainly on single-face detection [5, 13, 44, 57, 70, 75, 79, 81] and overlook essential contextual information, such as the scene relationships, interactions, and motion consistency among faces. This can result in misclassifying manipulated faces, which in turn may distort the overall interpretation of an event. Although a few multi-face detection methods exist [41, 43, 47, 50, 81, 84], they generally do not assess frame-level complete multi-face detection performance. Moreover, these methods extract features based on the heuristics of individual researchers, limited to individual perceptions or assumptions. *In multi-face deepfake scenarios, understanding how human cognition identifies deepfake cues in group settings remains an open research question.*

Our approach and rationale. In our work, we tackle the challenges of multi-face deepfake detection through a novel human-centric approach. Unlike previous methods that depend on off-the-shelf classifiers or heuristics shaped by individual researchers' perceptions, we focus on examining detection cues derived from crowdsourced human studies. This approach provides fresh insights into multi-face deepfake detection. By aggregating crowdsourced annotations, we identify multiple social contextual cues humans rely on for detecting deepfakes. Building on these cues, we propose HICOM, a framework designed to effectively detect multiple deepfake faces in social settings.

HICOM is grounded in a key rationale: rather than relying on off-the-shelf classifiers or individual heuristics, we base our model on human cognition. This is crucial because AI generation techniques evolve rapidly, and methods tailored to specific deepfake types often struggle to adapt to new variations. In contrast, human cognitive patterns remain stable, providing a reliable foundation for detection. Our rationale is supported by several studies in social science and neuroscience. First, research in social science shows that social context—capturing relationships within a group—plays a vital role in identifying inconsistencies [23]. Additionally, neuropsychological studies reveal that the human visual system is highly attuned to face perception [19, 34, 35]. We hypothesize that this sensitivity gives humans a natural advantage in detecting deepfake faces,

as they can instinctively recognize fake faces that don't fit within a social group [22]. Motivated by these insights, we explore how humans detect multi-face deepfakes in social settings and believe that incorporating human cognition can lead to effective and robust multi-face detection models.

As shown in Fig. 1, our human study reveals four key insights into the specific cues that frequently appear in multi-face deepfake scenarios, namely scene-motion coherence, inter-face appearance compatibility, interpersonal gaze alignment, and face-body consistency. Leveraging human insights, we identify key cognitive strategies used to detect fake faces in group contexts. These insights inform the development of our human-inspired, context-aware, multi-face deepfake detection framework, named HICOM. The framework consists of four modules: scene-motion module (M1), inter-face appearance module (M2), gaze module (M3), and body-face module (M4), with the weights of each module motivated by our human studies. Our contributions are as follows.

- **Human-inspired deepfake detection.** We pioneer the use of human studies to explore contextual features for multi-face deepfake detection in social settings. Our work introduces a novel analytical perspective and identifies key detection cues through a series of empirical studies. Insights from these studies inform the design of HICOM, which is tailored to align with how humans naturally detect deepfakes. This approach enhances detection interpretability and strengthens the persuasiveness of the results.
- **Human cognition on deepfake perception.** We examine human cognitive patterns in identifying fake faces in multi-face deepfake videos, identifying four key factors humans rely on during detection: scene-motion coherence, inter-face appearance compatibility, interpersonal gaze alignment, and face-body consistency. Our findings provide empirical insights for multi-face deepfake detection depicting natural social interactions.
- **Frame-level complete multi-face detection benchmark.** We emphasize the importance of detecting all fake faces in multi-face scenarios to enhance frame-level complete multi-face detection performance, which sets itself apart from existing single-face benchmarks. We argue that detecting deepfakes in social settings is crucial and propose a human-centered paradigm to address this challenge. We hope this pioneering research will highlight the importance of deepfake detection in social contexts and the vital role of understanding human cognition in combating this growing threat.

2. Related Work

Multi-Face Deepfake Generation. Understanding multi-face deepfake generation is essential for effective detection. Recent advancements include OpenForensics [39], FFIW

[84], ManualFake [26], and DF-Platter [53]. OpenForensics assesses manipulation feasibility using GAN-based synthesis [59, 67] and Poisson blending. FFIW automates multi-face swapping with tools like DFL [58] and FSGAN [55]. ManualFake uses commercial software for synthesis and involves DFL, FSGAN, and Simswap [11] manipulations, while DF-Platter generates a large-scale multi-face deepfake dataset using FSGAN [55] and FaceShifter [40]. However, these methods often overlook the social context of the faces, causing inconsistencies. Our human study highlights the importance of social context, which we incorporate into our multi-face deepfake detection framework.

Multi-Face Deepfake Detection. Most deepfake detection methods focus on single-face scenarios and fall into implicit clue-based [2, 7, 32, 48, 57, 79], signal clue-based [5, 14, 42, 45, 60], and semantic clue-based approaches [13, 25, 27, 32, 52, 70, 75]. These methods struggle in multi-face scenarios by ignoring contextual information and face relationships.

A few recent research has focused on multi-face deepfake detection. Limited works in this area include S-MIL [41], Zhou et al. [84], Ma et al. [47], FILTER [43], COMISC [81], and MoNFAP [50]. S-MIL employs sharp multiple instance learning for video-level multi-face detection, while Zhou et al. use a discriminative attention model for the same purpose. COMISC utilizes bi-grained contrastive learning, and MoNFAP uses noise extracts for detection, and FILTER focuses on extracting facial aggregation features, and Ma et al. use a VGG network to detect fake frames.

While these methods achieve promising performance, they typically rely on black-box classifiers or individual heuristics, and often lack evaluation of frame-level complete multi-face detection. Our work leverages human cognitive insights from multiple observers to significantly improve frame-level complete multi-face detection performance, enabling a more effective defense against deepfake threats.

Human Sensitivity in Face Perception. Neuroscience research has found that humans have dedicated neurobiological mechanisms for face recognition, primarily in the fusiform face area (FFA) and superior temporal sulcus [28, 36]. Human sensitivity in face perception plays a crucial role in social interactions and deception detection. Studies have shown that humans can rapidly recognize faces and detect subtle anomalies, such as unnatural textures or inconsistencies in expressions, which are often associated with deepfake or manipulated images [20, 34]. However, sensitivity to fake or abnormal faces varies based on context, prior exposure, and individual cognitive biases [18, 54]. While humans exhibit a general ability to detect manipulated faces, well-crafted deepfakes can still bypass perceptual defenses, leading to misjudgment [49]. Our

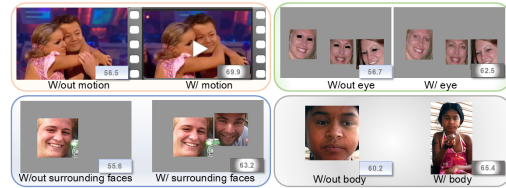


Figure 2. Examples of human studies with and without the four contextual features. The boxes display human performance in frame-level multi-face detection accuracy (%).

work is motivated by the above. We believe that understanding human perceptual mechanisms is essential for improving AI-driven deepfake detection and designing more effective countermeasures.

Context-Aware Modeling of Human Groups. In addition to their innate sensitivity to faces, another characteristic of humans is their tendency to interact in groups. Research in this area focuses on understanding various group attributes, including activities, age, and gender. In group activity recognition, researchers develop dynamic inference networks to analyze relationships among individuals. Notable advancements include graph network [78, 80], Dual-path Actor Interaction framework with Multi-scale Actor Contrastive Loss [23], and methods aligning local and global spatio-temporal views [10]. For age and gender detection in groups, contextual features prove beneficial. Gallagher et al. demonstrate that these features enhance age and gender prediction [23], while Rodriguez et al. introduce a feed-forward attention mechanism to improve age recognition in group images [61].

Inspired by these studies, we highlight the importance of social contextual cues within groups for detection. We examine how humans detect these cues and integrate them into our detection framework.

3. How Humans Detect Multi-Face Deepfake

3.1. Research Hypotheses

Inspired by prior research on face perception and social scene understanding, we believe that incorporating human cognitive characteristics can enhance deepfake detection models, which motivates our human study. In this subsection, we outline our research hypotheses for the human study and provide the rationale behind each.

Firstly, face replacement in multi-face deepfake videos often introduces scene-motion inconsistencies, disrupting the natural scene arrangement and motion coherence among individuals [3, 71]. Such inconsistencies appear as unnatural movements, jitter, or misalignments between faces and their surrounding context. Since humans naturally rely on scene coherence and motion smoothness to interpret group interactions [33, 66], we propose our first hypothesis:

H1: Deepfake techniques introduce scene-motion incoherence, which humans can identify as a key factor in deepfake detection.

Secondly, even with post-processing, deepfake faces often exhibit mismatches among faces, blending artifacts, or illumination inconsistencies within multi-face scenarios [17, 81]. Such discrepancies can create an unnatural appearance when a deepfaked face is compared to authentic faces in the same scene. Crowd analysis studies also find that inter-face appearance features are fundamental for understanding groups of people [65, 74]. Therefore, we propose our second hypothesis:

H2: Deepfake faces exhibit inter-face appearance inconsistencies in resolution, color, or illumination in the scene, serving as contextual cues for human detection.

Thirdly, human gaze direction is a critical factor in both visual saliency and social perception. Research in gaze and psychology indicates that gaze plays an essential role in group settings [31, 82]. Studies have shown that gaze alignment is fundamental to social interactions, influencing attention and trustworthiness judgments [38, 64]. Deepfake synthesis often fails to maintain natural gaze consistency, resulting in mismatches between the faked face and others in the scene [8, 46]. Building on these findings, we propose the third hypothesis for our human study:

H3: Inconsistencies in gaze direction between deepfake faces and other individuals in a multi-face scene will be a detectable cue for humans.

Lastly, deepfake generation methods often overlook body-face coherence, as most models focus primarily on facial synthesis rather than holistic body alignment [9, 83]. This lack of contextual awareness can lead to discrepancies between the generated face and body, particularly in terms of age and gender. Research on human behavior in groups has shown that age and gender are crucial factors for autonomous detection [16, 76]. Based on this, we propose the fourth hypothesis:

H4: Deepfake faces may show inconsistencies in body age and gender, providing an additional cue for detection.

3.2. Human Study

Based on our research hypotheses, we conduct a two-phase human study to explore human detection of multi-face deepfakes. In the first phase, we randomly selected 2,000 multi-face deepfake videos and images from the OpenForensics [39], FFIW [84], and DF-Platter [53], with each video lasting approximately 20 seconds. These datasets are the available benchmarks of current multi-face deepfakes. Four university students were recruited for this phase. Each participant was assigned 500 videos and images to review. They were compensated at \$10 per hour.

Participants documented the fake faces they identified, noted their reasons. They reviewed images and videos di-

rectly via a PC media player, without needing frame-by-frame analysis. Only identifications matching the dataset labels were considered valid. Participants categorized the 500 samples according to the detection cues, so that we can calculate the prevalence of each cue across all samples.

The first phase identified 8 primary indicators, summarized in Fig. 1. These were distilled into 4 hypotheses for multi-face deepfake detection: scene-temporal artifacts (H1), inter-face appearance anomalies (H2), gaze direction inconsistencies (H3), and mismatches between body and face movements (H4). Using these findings, we designed the second phase of our study to explore how these cues influence detection accuracy.

In the second phase, we sampled an additional 920 videos and images from our dataset pool and manipulated the fake faces across various scenarios to examine the impact of multiple contextual cues. These scenarios—blocked motions, blocked surrounding faces, blocked eyes, and blocked bodies—each isolates one of the four key contextual cues identified in the first phase (as shown in Fig. 2). We recruited 20 participants from the online crowdsourcing platform Amazon Mechanical Turk [15] to identify the cues. To ensure reliability, participants are selected based on their approval rate and demographic suitability. Results in Fig. 2 demonstrate that performance improves when incorporating these contextual features, highlighting their significance in human detection.

3.3. Human Cues in Detecting Deepfakes

As shown in Fig. 1, four types of factors were identified to assist in human detection, with an emphasis on contextual elements. Minor cues (1.8%) like background-text consistency were excluded from our integration. The most common cue, accounting for 34.2%, stems from discontinuous motions between preceding and following frames, supporting H1. Additional significant factors, together representing 31.5%, include inconsistencies that emerge in the inter-face context. These encompass variations in face resolution, mismatched lighting, color inconsistencies, and artifacts that appear in some faces but not others, supporting H2. The context of gaze within groups also plays a crucial role, with abnormal gaze direction relative to the camera accounting for 25.0% of the cases (H3). Lastly, within the context of body and face alignment, discrepancies in age and gender between faces and bodies contribute 7.5% to detection difficulties (H4).

4. Context-Aware Multi-Face Detection

4.1. Overall Framework and Design Rationale

Leveraging insights from our human study, we propose a context-aware multi-face deepfake detection method that integrates human-derived reasons. Unlike methods that

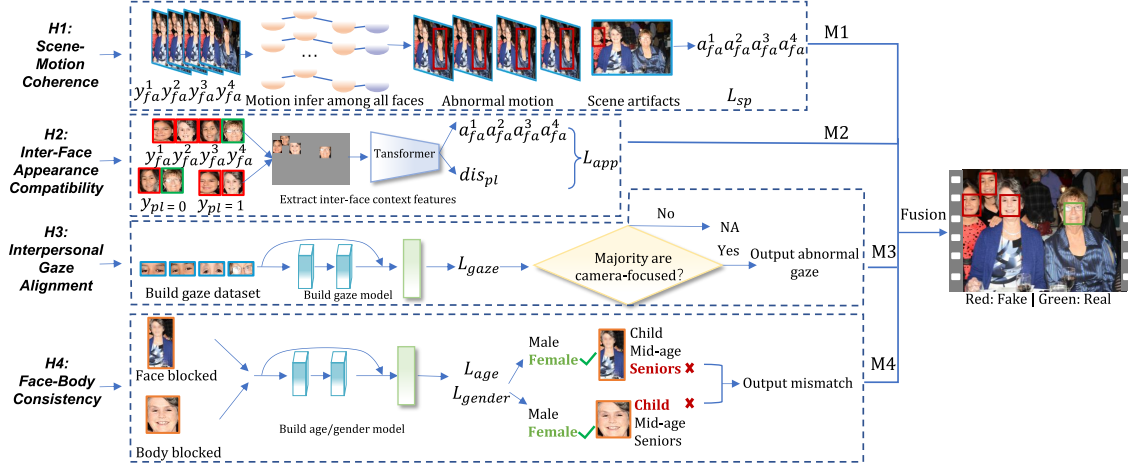


Figure 3. HICOM leverages human-inspired cues ($H1 - H4$) derived from human studies to detect all fake faces within multi-face settings.

merely superimpose models, each module of HICOM is inspired by specific human-reported cues. According to the $H1$ (scene-motion coherence), we develop a module that integrates facial and contextual features from preceding and following frames to expose unnatural motion and scene inconsistencies. We address the $H2$ (inter-face appearance compatibility) by developing an inter-face appearance module. This module enhances detection performance by combining single-face classification with multi-face comparisons. Inspired by $H3$ (interpersonal gaze alignment), we devise a gaze module that isolates eye regions to model gaze behavior, identifying anomalies such as a single eye not aligning with others directed toward the camera. Finally, we utilize $H4$ (face-body consistency) to devise a body-face module that independently assesses the age and gender of the face and body, detecting inconsistencies between them.

As illustrated in Fig. 3, HICOM comprises four modules ($M1-M4$), each inspired by specific human-cognitive insights. Rather than merely superimposing models, HICOM is grounded in human cognitive studies, with each module designed to integrate specific human-inspired contextual features. We leverage specialized feature extraction strategies, including inference from [80], Transformer networks [73], and ResNet architectures [29], tailored specifically for each module. Crucially, this modular design ensures robustness: even if three modules fail to detect anomalies, the remaining module can independently identify deepfake cues.

4.2. Scene-Motion Module

$H1$ identifies scene-motion inconsistency as the critical cue for deepfake detection, prompting us to develop a module that simulates this perception. The scene-motion module constructs multi-scale features by extracting detailed information from each face and its surrounding regions. By inferring motions and extracting scene inconsistencies across

facial and background context over time, this module effectively detects scene-motion inconsistencies in fake faces.

Multi-Scale Feature Extraction. We extract multi-scale features from sequences of images. The input data is reshaped to capture both scene and motion dimensions, allowing the network to generate comprehensive multi-scale feature representations across multiple frames. These features are refined using RoIAlign [30], which focuses on specific regions corresponding to detected faces and backgrounds. By embedding the extracted features through a fully connected layer, the model enhances its ability to detect scene-motion inconsistencies in multi-face scenarios.

Scene-Motion Inference. The scene-motion module infers motions and detects inconsistencies by analyzing each face and scene features across time. Inference network [80] is then used to focus on significant motion patterns, refining the detection of scene-motion inconsistencies. By combining features across time, the model generates a comprehensive representation that enhances detection accuracy.

The output features are processed to predict face-level and frame-level complete multi-face detection scores, allowing the model to identify deepfake manipulations effectively. Cross-entropy loss optimizes the model during training, ensuring robust detection across multiple faces.

$$\mathcal{L}_{sp} = \lambda_{fa} \text{CE}(a_{fa}, y_{fa}) + \lambda_{fr} \text{CE}(a_{fr}, y_{fr}), \quad (1)$$

where $\text{CE}(a_{fa}, y_{fa})$ represents the face-level cross entropy loss for faces a_{fa} with the true label y_{fa} , and λ_{fa} and λ_{fr} represents the equal weight for the face-level loss and frame-level loss, and $\text{CE}(a_{fr}, y_{fr})$ represents the frame-level cross entropy loss for frames a_{fr} with the true label y_{fr} , which checks if all faces in the frame are correctly predicted.

4.3. Inter-Face Appearance Module

$H2$ underscores the importance of inter-face context in deepfake detection, prompting the design of an inter-face appearance module. The inter-face appearance module focuses on face regions

and the comparisons of multi-faces in a frame. By extracting facial features and comparing different faces, inter-face appearance module can detect the inconsistency among faces.

Inter-Face Comparisons. We crop face regions and use the aforementioned Transformer for model building. To compare multi-face features, we combine a contrastive loss and cross entropy loss for training.

$$\mathcal{L}_{app} = \text{CE}(a_{fa}, y_{fa}) + \frac{\lambda_{comp}}{N_{comp}} \sum_{j=1}^{N_{comp}} \left[y_{pl}^j \cdot dis_{pl}^j + (1 - y_{pl}^j) \cdot \max(0, \text{margin} - dis_{pl}^j) \right], \quad (2)$$

where $\mathcal{L}_{app}$ denotes the total combined loss, N_{comp} denotes the number of pair of samples, y_{pl}^j denotes the binary label for the j -th pair of faces (where $y_{pl}^j = 1$ if the faces are with similar label and $y_{pl}^j = 0$ if they are dissimilar), dis_{pl}^j denotes the Euclidean distance between the feature vectors of pair of faces, margin denotes the minimum distance required for dissimilar pairs and is set to a default value of 1.0, λ_{comp} is empirically set as 0.3.

4.4. Gaze Module

Our human study and previous work [37, 85] show that outlier observers often have gaze points that do not align with the group's common gaze, yet these outliers are not necessarily fake. To reduce false positives, the gaze module identifies abnormal gazes by analyzing eye regions to determine if the gaze is locked on the camera. According to H3, we design a gaze module that filters out multi-face videos and images where most faces are not looking at the camera. It then detects abnormal gazes by checking for camera-focused gazes. Building a trained gaze-locking model, we apply a decision strategy to identify abnormal gazes.

Gaze Locking Model Construction. We crop eye regions from sequences of images or frames and label them based on whether the gaze is directed at the camera. To expand our dataset, we use the Columbia Gaze-DataSet [69], which includes data with diverse head poses and gaze directions, providing robust data for training. We then pretrain a model using the Columbia Gaze-DataSet and a Resnet for gaze classification. Thereafter, we use our built dataset to train the Resnet. The model is optimized with cross-entropy loss ($\mathcal{L}_{gaze}$) calculated from ground-truth and predicted gaze labels and is saved when validation loss converges.

Gaze Abnormal Detection. Not all faces with outlier gazes are fake [12, 24]. The module disregards multi-face images and videos where most faces are not looking at the camera. A face is flagged as fake only if most faces in the frame are looking at the camera while a few outliers are not. This is defined as:

$$a_i = \begin{cases} NA, & \text{if majority faces not looking at camera,} \\ 1, & \text{if } a_i \text{ is off-camera and } (n_L - n_O > 1 \text{ or } n_T = 2), \\ 0, & \text{if } a_i \text{ is not off-camera.} \end{cases} \quad (3)$$

where n_L , n_O , and n_T represent the number of faces looking at the camera, not looking at the camera, the total number of faces in the image or frame.

4.5. Body-Face Module

H4 underscores the importance of body-face context in deepfake detection, motivating the design of a dedicated body-face mod-

ule. This module detects mismatches between the face and body in terms of age and gender.

Face Block and Body Block. Body-blocked regions emphasize facial appearance, while face-blocked regions highlight clothing and posture features. To isolate body features for age and gender modeling, we apply GaussianBlur [21] to block the face areas within each body region. For face-only modeling, we crop faces from the images or frames to block body regions effectively.

Age & Gender Model Construction. For age model training, we categorize cropped faces and preprocessed body images into three groups: child, middle-aged, and senior. For gender model training, we classify them as male or female. Using the IMDB-WIKI dataset [63], we train Resnet to obtain a trained age and gender model, ensuring high age/gender detection performance. We then use this pretrained model to extract age and gender features.

Mismatch Detection. We optimize the age and gender models with cross-entropy losses ($\mathcal{L}_{age}$ and $\mathcal{L}_{gender}$) and save the models upon convergence. Let ag_i^{body} denote the predicted age or gender for the i -th body corresponding to the face, and ag_i^{face} denote the predicted age or gender for the i -th face corresponding to the body. Detection is determined by:

$$a_i = \begin{cases} 1, & \text{if } (ag_i^{face}) \neq (ag_i^{body}), \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Effects of the Module. Not all multi-face video images exhibit detectable body-face mismatches in age and gender. Therefore, this module acts as an auxiliary to other modules. When other modules fail to detect all fake faces in multi-face images, this module's results can supplement them, improving frame-level complete multi-face detection performance.

4.6. Module Combination

Since our human study shows that cues from M1 and M2 are more significant than those of M3 and M4 for detection, M1 and M2 serve as the primary components in our framework. In contrast, M3 and M4 are designed to provide complementary support. When M1 and M2 miss a fake face, M3 and M4 help identify these inconsistencies, thereby enhancing frame-level complete multi-face detection performance. Inspired by these insights and previous literature [4, 51], we fuse the outputs of these modules using an XOR operation, ensuring that any detected anomaly leads to a fake face prediction. The effectiveness of this fusion strategy is discussed further in the Supplementary Material.

5. Evaluation of Multi-Face Detection

5.1. Experimental Settings

Datasets. We conduct experiments using four benchmark multi-face deepfake datasets: FFIW [84], OpenForensics [39], DF-Platter [53], and ManualFake [26], which are all widely-used benchmark datasets on multi-face deepfake. FFIW is a real-world multi-face deepfake video dataset, with frames containing up to 15 faces. OpenForensics comprises GAN-generated images with an average of 2.9 faces per image. Since OpenForensics is image-based, we replicate each image to create a sequence for input into the M1. DF-Platter is a multi-face deepfake video dataset with 2-5 faces per video. ManualFake provides multi-face deepfake

Method	FFIW				OpenForensics				DF-Platter			
	FAC	FAU	FCAC	FCAU	FAC	FAU	FCAC	FCAU	FAC	FAU	FCAC	FCAU
SBI* [68]	94.0	94.2	84.1	85.6	92.8 [†]	98.8 [†]	83.4	85.7	95.7	96.3	88.7	88.9
TALL* [75]	94.6	95.5	88.9	89.7	98.2	98.4	93.1	94.6	96.8	96.9	90.2	91.7
Li et al.* [44]	86.3	91.1	77.2	78.9	91.1	93.4	80.7	81.9	93.9	95.0	89.2	89.5
Zhou et al. [84]	85.4	85.9	72.3	73.6	93.2	94.8	86.5	87.8	90.4	91.6	80.4	80.6
Ma et al. [47]	88.4	91.5	82.5	83.2	96.4	98.5	87.6	88.2	95.2	96.5	89.2	89.9
FILTER [43]	92.5	94.4	84.9	85.4	99.0 [†]	99.9[†]	93.6	93.7	96.8	97.5	89.5	90.6
MoNFAP [50]	91.7	94.3	80.2	82.1	99.1	99.9[†]	89.6	92.3	92.6	93.7	88.4	89.3
COMISC [81]	93.2	94.7	85.0	85.6	98.4	99.5	93.7	94.8	93.4	94.8	89.2	89.7
HICOM	94.7	95.9	91.3	92.1	99.3	99.9	97.8	98.9	97.2	98.4	93.5	94.6

Table 1. Comparisons of in-dataset detection performance between HICOM and other methods on multi-face datasets. For all tables, results marked with [†] are cited from FILTER [43]. Single-face methods are denoted by *, while multi-face methods are unmarked.

versions transmitted through online social networks. Following MoNFAP [50], we use ManualFake to evaluate generalization in untrained real-world scenarios.

Implementation Details. We use the Adam optimizer with an initial learning rate of 1×10^{-4} , training for 120 epochs, and applying a decay rate of 1/3 every 10 epochs. The size of M1 is 720×1280 , while other modules use a size of 224×224 . Experiments are conducted on NVIDIA H100 80GB GPUs.

Metrics. We report face-level ACC (FAC), face-level AUC (FAU), frame-level complete multi-face detection ACC (FCAC), and frame-level complete multi-face detection AUC (FCAU) scores. Face-level metrics assess each face independently, while frame-level complete multi-face detection metrics evaluate the detection of each face within that frame.

Baselines. We compare HICOM with representative single-face detection methods: SBI [68], TALL [79], and Li et al. [44], as well as the limited number of recently published SOTA multi-face detection methods, including Zhou et al. [84], Ma et al. [47], FILTER [43], MoNFAP [50], and COMISC [81].

5.2. In-Dataset Detection Performance.

We conduct in-dataset experiments on FFIW, OpenForensics, and DF-Platter, using the same datasets for both training and testing. As shown in Table 1, while both single-face and multi-face detection methods perform well in face-level metrics, they degrade in frame-level complete multi-face detection metrics. However, HICOM achieves average improvements of 3.3% in FCAC, and 3.1% in FCAU compared to the next best results. This success stems from the method’s thorough consideration of contextual features, including scene-motion coherence, inter-face appearance compatibility, interpersonal gaze alignment, and face-body consistency in terms of age and gender, ensuring comprehensive detection and minimizing missed fakes.

5.3. Model Generalizability

Robustness to Unseen Real-World Perturbations. Real-life deepfakes often involve various perturbations, and OpenForensics simulates this by providing six types: color manipulation, edge manipulation, block-wise distortion, image corruption, convolution mask transformation, and external effects. To assess HICOM’s robustness to these unseen perturbations, we conduct

Method	OpenForensics with Perturbations			
	FAC	FAU	FCAC	FCAU
SBI* [68]	74.7 [†]	82.5 [†]	66.1	67.4
TALL* [75]	90.7	96.5	77.1	78.4
Li et al.* [44]	75.6	75.8	63.7	64.9
Zhou et al. [84]	78.9	79.5	64.7	68.9
Ma et al. [47]	78.4	81.6	63.6	63.9
FILTER [43]	89.0 [†]	96.9 [†]	74.3	76.8
MoNFAP [50]	87.3	89.2	72.8	74.9
COMISC [81]	88.2	92.1	73.0	74.5
HICOM	91.2	97.5	78.6	81.2

Table 2. Robustness comparisons in unseen perturbations.

experiments on OpenForensics, where none of the perturbations were included in the training process. Results in Table 2 show that while existing methods struggle to generalize to unseen perturbations, HICOM achieves an average improvement of 1.5% FCAC and 2.8% FCAU over the previous best results. This improvement stems from HICOM’s reliance on contextual features, which are less dependent on specific training data and more resilient to perturbations. For example, abnormal gaze and mismatches in age and gender between faces and bodies, identified during training, remain detectable even under perturbations in the test set. Additional experiments on videos with unknown compression factors are detailed in the Supplementary Material.

Generalization to Unseen Dataset. To evaluate the generalization of the identified cues, we conduct cross-dataset experiments, training the model on DF-Platter or FFIW and testing it on ManualFake. Notably, ManualFake was not used for human studies or model training, and OpenForensics was excluded from training as it contains only images rather than videos. The results presented in Table 3 indicate that all state-of-the-art methods exhibit a significant drop in performance, highlighting the substantial challenge of multi-face deepfake detection. Nevertheless, HICOM achieves an average improvement of 5.8% at the frame-level complete multi-face detection accuracy, demonstrating that the cues inspired by human studies and incorporated into our design exhibit a certain degree of generalization.

Single-Face Detection. Our method can be adapted to single-

Method	DF-Platter to ManualFake				FFIW to ManualFake			
	FAC	FAU	FCAC	FCAU	FAC	FAU	FCAC	FCAU
SBI*[68]	69.1	70.7	59.3	60.9	70.3	71.4	63.3	63.9
TALL*[75]	69.3	70.4	59.9	60.6	71.3	72.2	65.8	66.7
Li et al.*[44]	68.3	69.4	56.6	57.5	69.7	69.9	58.1	59.3
Zhou et al. [84]	64.2	65.9	56.1	56.3	68.4	69.1	56.2	57.7
Ma et al. [47]	63.8	64.7	55.1	56.2	68.3	69.6	56.3	56.9
FILTER[43]	68.1	69.4	56.8	57.7	66.4	68.3	61.8	62.2
MoNFAP[50]	60.7	67.9	53.2	54.3	61.6	62.2	55.7	56.2
COMISC[81]	67.7	68.9	58.2	59.4	68.8	69.9	62.6	63.3
HICOM	70.7	71.4	66.3	67.7	72.8	73.3	70.9	71.6

Table 3. Generalization comparisons in untrained datasets.

Module	FFIW		OpenForencics		DF-Platter	
	FCAC	FCAU	FCAC	FCAU	FCAC	FCAU
M1	87.7	89.0	93.9	95.6	90.4	91.1
M1+M2	89.9	90.6	95.7	97.3	92.0	92.8
M1+M2+M3	90.6	91.4	97.2	98.3	93.3	94.0
M1+M2+M3+M4	91.3	92.1	97.8	98.9	93.5	94.6

Table 4. Ablation study - Comparisons of frame-level complete multi-face detection performance in different modules.

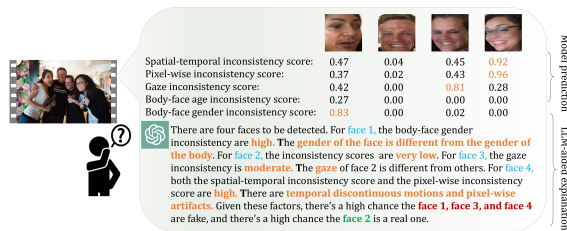


Figure 4. HICOM provides comprehensible explanations for its predictions through integration with an LLM.

face scenarios. Specifically, we modify M1 to extract only scene-motion features and M2 to focus solely on single-face features, removing inter-face dependencies. M3 is excluded as it is not applicable, while M4 remains unchanged. We evaluate this adaptation on the FF++ [62] dataset, demonstrating competitive performance against SOTA methods in single-face detection. Due to space constraints, detailed results are provided in Supplementary Material.

5.4. Analyses and Discussions

Effects of Four Modules. We progressively evaluate modules M1 through M4, as shown in Table 4. M1 alone achieves acceptable performance, while adding subsequent modules consistently improves results. M2 significantly boosts accuracy by combining classification and contrastive loss across multiple faces. M3 provides modest gains by targeting gaze anomalies but is less effective when gazes are dispersed. M4 contributes the least, as it only activates for specific face-body inconsistencies. Nonetheless, M3 and M4 remain essential for detecting faces missed by earlier modules.

LLM Explanation. We use the output scores of M1, M2, M3, and M4 as prompts and invoke the ChatGPT API [1] to explain

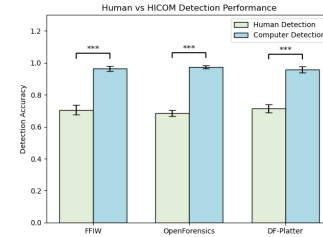


Figure 5. HICOM surpasses humans in multi-face detection.

HICOM, which shows our model’s potential to work with LLM to provide an explainable prediction. Results in Fig. 4 show that each face is detected and explained as either fake or real, enhancing user trust and understanding of the detection.

Human Detection Results. We conduct human studies on multi-face detection using 300 randomly selected samples from FFIW, OpenForencics, and DF-Platter, comparing the results with HICOM. Each set is evaluated by 5 AMT workers, with their averaged performance and standard error reported. We label a p -value less than 0.0005 with three stars, a p -value between 0.0005 and 0.005 with two stars, and a p -value greater than 0.005 with one star. Results in Fig. 5 show three stars for the p -value, indicating that HICOM outperforms human detection. This demonstrates its effectiveness in assisting users with multi-face deepfake detection.

6. Conclusion

This paper presents a novel framework that leverages crowd-sourced human studies, approved by the university’s Institutional Review Board (IRB), to systematically detect every single fake face in multi-face scenarios. Emphasizing cognitive processes as fundamental to deepfake detection, the proposed framework HICOM moves beyond traditional methods that rely solely on black-box classifiers or individual heuristics. Instead, it integrates human cognitive insights derived from multiple observers into the detection framework. Quantitative results identify scene-motion coherence, inter-face appearance compatibility, interpersonal gaze alignment, and face-body consistency as key factors in multi-face deepfake detection. By incorporating these human-inspired cues, HICOM demonstrates how social context provides a richer context for distinguishing real from fake faces in group settings. This work represents a pioneering step in detecting multiple fake faces within social contexts.

Limitations. Our findings are based on all benchmark multi-face deepfake datasets, with cues specifically designed for fake face detection. As a result, some cues (e.g., gaze alignment and face-body consistency) may not universally apply to natural images. Additionally, as deepfake techniques evolve, new contextual cues may emerge. However, our paradigm and hypotheses are grounded in fundamental human cognitive patterns, making them broadly applicable to future deepfakes.

While our modular approach offers flexibility and generalizability in complex multi-face scenarios, it is not optimized for end-to-end settings. Nevertheless, given the diverse manipulation cues across multiple faces, this design remains advantageous for accurate detection and interpretation. Moreover, it can be easily adapted to incorporate new cues as deepfake techniques evolve.

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